

MACHINE LEARNING (ML)

4 Types

- ↳ Supervised learning
- ↳ Unsupervised learning
- ↳ Semi-supervised learning
- ↳ Reinforcement learning

① ONLINE LEARNING :-

→ Datasets that arrive continuously and it can be trained without putting down the server.

Working :-

- ① Receives new data.
- ② Update Model
- ③ Make prediction
- ④ Repeat.

App Recommendation Systems

② BATCH LEARNING :-

ML is trained on the entire dataset at once.

Working :-

- ① Collect data.
- ② Train model on all data.
- ③ Deploy Model.
- ④ Retrain when new data arrives.

App Neural Networks

Instance-Based Learning

- Stores up.
- Lazy learning
- Fast training
- Slow prediction
- High Memory

Model Based Learning

- learns formula
- eager learning
- Slow training
- fast prediction
- Low memory

TENSORS

→ A tensor is multidimensional array used to represent data in ML and DL.

Tensors = Generalization of scalars, vectors, and matrices.

↳ 0D → 5

↳ 1D → [1, 2, 3]

↳ 2D → [[1, 2, 3], [4, 5, 6]]

↳ 3D → RGB Image

↳ (3, 224, 224)

↳ 4D → (32, 224, 224, 3)

How to frame a ML problem

Business Problem to
ML Problem.

↓
Types of Problem

↓
Current Solⁿ.

↓
Getting Data.

↓
Metrics to Measure

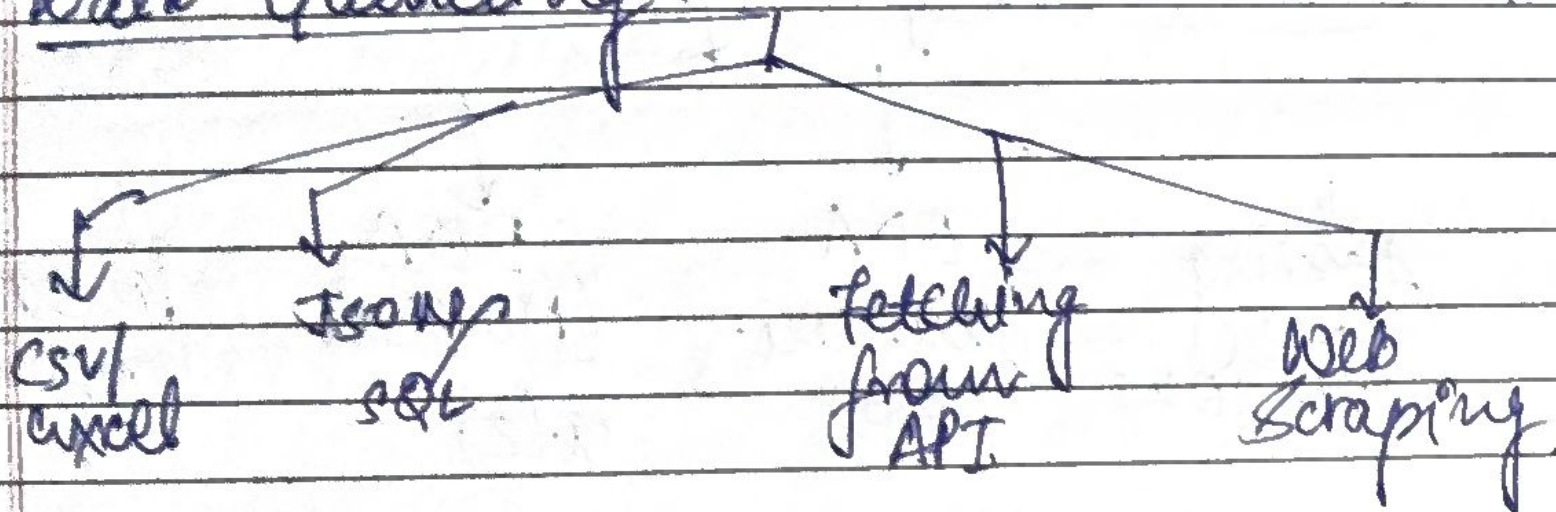
↓
Online vs Batch

↓
Check Assumption.

App

Predict the
Customer about
to Churn from
Netflix

Data Gathering



→ CSV/Excel - used when data is already available as files

→ JSON

Used for semi-structured data, mostly from APIs.

→ SQL

Used when data is stored in databases.

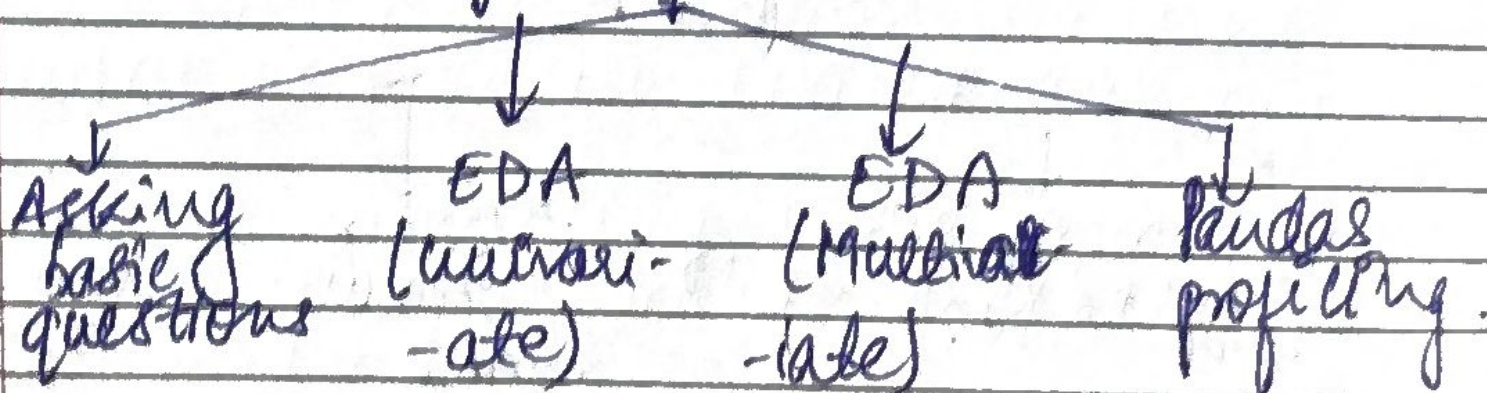
→ APIs

Used to fetch real time, or live data.

→ Web Scraping

Used only when no APIs is available.

Understand your data



① Asking basic Questions

- How big is your data? → `df.shape`.
- How does the data look like? → `df.head()`.

- What is the data type of cols? $\rightarrow$ `df.info()`.
- Are there any missing values? $\rightarrow$ `df.isnull().sum()`.
- How does the data look mathematically? $\rightarrow$ `df.describe()`.
- Are there duplicate values? $\rightarrow$ `df.duplicate.sum()`.
- How is the correlation b/w columns? $\rightarrow$ `df.corr()`.

② EDA (univariate)

- $\rightarrow$ One variable at a time. Basically how does this single column look like?
- $\rightarrow$ Check: Distribution, Central tendency, Outlier, Missing Values.
- $\rightarrow$ Common Visuals: Histogram, Box plot.

③ EDA (bivariate)

- $\rightarrow$ Two variables together
- $\rightarrow$ How one column change with another?
- $\rightarrow$ Check: Correlation, Trends
- $\rightarrow$ Common Visuals: Scatter plot.

Multivariate

- More than 2 variable.
- How do multiple columns interact together.
- Check Combined effects, patterns.
- Common Multivariate Heatmaps, pair plot.

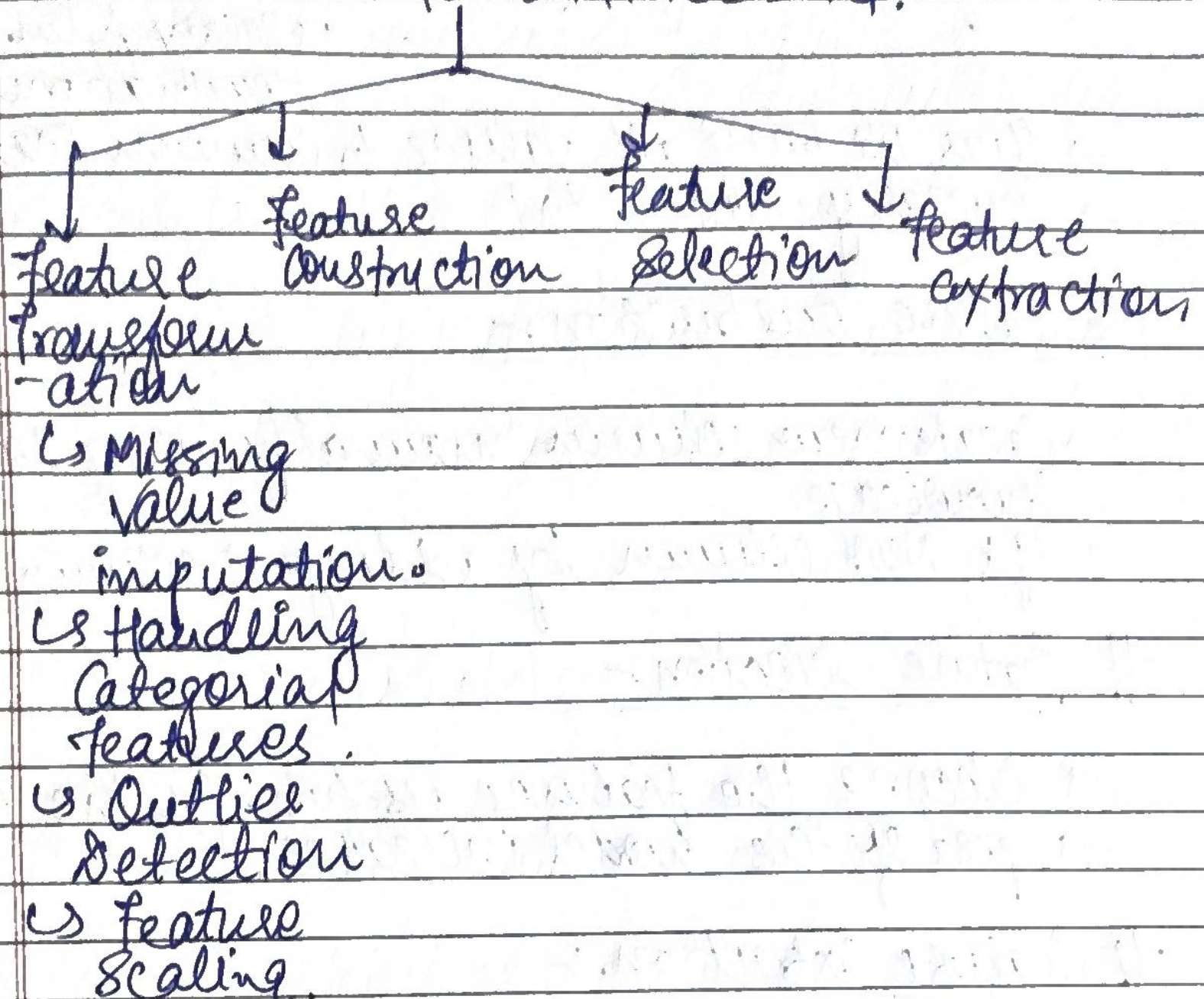
④ Panda's profiling

- An automated EDA tool that summarizes the dataset for you.
- Helps you quickly understand structure & quality of data.
- Useful for initial exploration, not final conclusions.

WHAT IS FEATURE ENGINEERING?

- Feature engineering is the process of using domain knowledge to extract features from raw data.
- These features can be used to improve the performance of machine learning algorithms.

FEATURE ENGINEERING.



Feature Transformation

① Missing Value Imputation :-

Either remove or fill missing values. Replace by mean, median or most frequent category.

② Handling Categorical features :- e.g. One-hot encoding.

③ Outlier Detection - Outliers removed are much better fit.

④ Feature Scaling - e.g. Age in range of 100 & Salary in range of 90000 &

→ So for KNN eq → $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 ↳ salary form
 denominations.

→ Thus we bring all values to common scale
 → Generally b/w -1 to 1

Feature Construction

→ Create new column manually using domain knowledge.
 → Cgr New column by adding 2 column.

Feature Selection

→ selecting the features useful for the model.
 → speed of the model increases.

Feature Extraction

→ egr PCA (Principal Component Analysis), LDA
 → This is done by system, not manually.